

Homework assignment 1

1. First order logic, conjunctive normal form

Considering that:

- Every project is related to at least one research topic.
- For any project, any student working on that project studies every topic related to that project.
- If someone is supervising a project, then there must be a student who works on that project.
- Cris works on proj1.
- AI and Logic are topics of proj2
- Alice supervises proj2.

1.1. Part A: convert the statements to first-order logic:

Let:

P = project
 T = research topic
 S = student
 $related(T, P)$ = T is related to P
 $works(S, P)$ = S works on P
 $studies(S, T)$ = S studies T
 $supervises(S, P)$ = S supervises P

“Every project is related to at least one research topic”:

$$\forall P. \exists T. related(T, P)$$

“For any project, any student working on that project studies every topic related to that project”:

$$\forall P. \forall S. \forall T \left((works(S, P) \wedge related(T, P)) \rightarrow studies(S, T) \right)$$

“If someone is supervising a project, then there must be a student who works on that project”:

$$\forall P. (\exists X. supervises(X, P) \rightarrow \exists S. works(S, P))$$

“Cris works on proj1”:

$$works(cris, proj1)$$

“AI and Logic are topics of proj2”:

$$related(ai, proj2) \wedge related(logic, proj2)$$

“Alice supervises proj2”:

$$supervises(alice, proj2)$$

1.2. Part B: Conversion to CNF

$$\begin{aligned} & \forall P. \exists T. related(T, P) \\ \forall P. related(f(P), P), & \quad \text{skolemisation (T becomes the skolem function } f(P)) \\ related(f(P), P), & \quad \text{generalisation} \\ \therefore \boxed{C_1 = \{related(f(P), P)\}} \end{aligned}$$

$$\begin{aligned} & \forall P. \forall S. \forall T. ((works(S, P) \wedge related(T, P)) \rightarrow studies(S, T)) \\ \forall P. \forall S. \forall T. (\neg(works(S, P) \wedge related(T, P)) \vee studies(S, T)), & \quad \text{implication removal} \\ \forall P. \forall S. \forall T. ((\neg works(S, P) \vee \neg related(T, P)) \vee studies(S, T)), & \quad \text{De Morgan} \\ (\neg works(S, P) \vee \neg related(T, P)) \vee studies(S, T), & \quad \text{generalisation} \\ \therefore \boxed{C_2 = \{\neg works(S, P), \neg related(T, P), studies(S, T)\}} \end{aligned}$$

$$\begin{aligned} & \forall P. (\exists X. supervises(X, P) \rightarrow \exists S. works(S, P)) \\ \forall P. (\neg \exists X. supervises(X, P) \vee \exists S. works(S, P)) & \quad \text{implication removal} \\ \forall P. (\forall X. \neg supervises(X, P) \vee \exists S. works(S, P)) & \quad \text{pushing implication} \\ \forall P. \forall X. \exists S. (\neg supervises(X, P) \vee works(S, P)) & \quad \text{prenex form} \\ \forall P. \forall X. (\neg supervises(X, P) \vee works(g(P, X), P)) & \quad \text{skolemisation} \\ \neg supervises(X, P) \vee works(g(P, X), P) & \quad \text{generalisation} \\ \therefore \boxed{\{\neg supervises(X, P), works(g(P, X), P)\}} \end{aligned}$$

And we have the following facts:

$$\begin{aligned} & works(cris, proj1) \\ \therefore \boxed{C_4 = \{works(cris, proj1)\}} \\ & related(ai, proj2) \wedge related(logic, proj2) \\ \therefore \boxed{C_5 = \{related(ai, proj2)\}}, \quad \boxed{C_6 = \{related(logic, proj2)\}} \\ & supervises(alice, proj2) \\ \therefore \boxed{C_7 = \{supervises(alice, proj2)\}} \end{aligned}$$

2. Natural deduction

2.1. Natural deduction, and proposition

Use natural deduction to prove:

$$(A \vee B) \wedge \neg B \rightarrow A$$

Proof.

$$\frac{\frac{\frac{[(A \vee B) \wedge \neg B]^1}{[B]^2 \quad \neg B_{\wedge E}}{B \wedge \neg B_{\wedge I}} \quad \perp_{\perp I}}{[A]^3 \quad A \vee B \quad A_{\vee E^{2,3}}} \quad \underline{A}_{\rightarrow I^1}}{(A \vee B) \wedge \neg B \rightarrow A}$$

2.2. Natural deduction as proof for 1(a)

Prove that the following holds true, using the FOL statements from 1(a) as premises:

There is a research topic that Cris studies

Proof.

Premises and variable assignment; I can't include these in the tree; (it goes off-screen)

- $p_1 = works(cris, proj1)$
- $p_2 = \forall P. \exists T. related(T, P)$
- $p_3 = \forall P. \forall S. \forall T. (works(S, P) \wedge related(T, P) \rightarrow studies(S, T))$
- $s_1 = (works(cris, proj1) \wedge related(\alpha, proj1)) \rightarrow studies(cris, \alpha)$

Proof.

$$\frac{\frac{\frac{p_2_{\forall E}}{works(cris, proj1) \wedge related(\alpha, proj1)_{\wedge I}} \quad \frac{p_1}{studies(cris, \alpha)_{\rightarrow E}} \quad \frac{p_3}{\exists T. studies(cris, T)_{\exists I}}}{\exists T. related(T, proj1) \exists T. studies(cris, T)_{\exists E^1}} \quad \underline{\exists T. studies(cris, T)_{\exists I^1}}}{\exists T. studies(cris, T)}$$

Resolution

Use resolution to show that the following holds, using CNF from 1(b)

There is a student studying ai

Useful clauses:

- $C_1 = \{related(f(P), P)\}$
- $C_2 = \{\neg works(S, P), \neg related(T, P), studies(S, T)\}$
- $C_3 = \{\neg supervises(X, P), works(g(P, X), P)\}$
- ~~$C_4 = \{works(cris, proj1)\}$~~
- $C_5 = \{related(ai, proj2)\}$
- ~~$C_6 = \{related(logic, proj2)\}$~~
- $C_7 = \{supervises(alice, proj2)\}$

Consider the following unifications:

$$\begin{aligned}
 C_1 &= \{related(f(proj2), proj2)\} : [P \mapsto proj2] \\
 C_2 &= \{\neg works(S, proj2), \neg related(ai, proj2), studies(S, ai)\} : [P \mapsto proj2, T \mapsto ai] \\
 C_3 &= \{\neg supervises(alice, proj2), works(g(proj2, alice), proj2)\} : [X \mapsto alice, P \mapsto proj2]
 \end{aligned}$$

PBC: negate the goal:

$$\begin{aligned}
 &\exists S. studies(S, ai) \\
 &\neg \exists S. studies(S, ai) \text{ negation} \\
 &\equiv \forall S. \neg studies(S, ai) \text{ pushing negation} \\
 &\equiv \neg studies(S, ai) \text{ generalisation} \\
 &\therefore C_8 = \boxed{\neg studies(S, ai)} \text{ CNF form}
 \end{aligned}$$

Resolution:

$$\begin{array}{c}
 \frac{C_2 \ C_8}{\neg works(S, proj2), \neg related(ai, proj2)} \text{resolvent: } studies(S, ai) \\
 \frac{\neg works(S, proj2), \neg related(ai, proj2) \ C_5}{\neg works(g(proj2, alice), proj2)} \text{resolvent: } related(ai, proj2) \\
 \frac{\neg works(g(proj2, alice), proj2) \ C_3}{\neg supervises(alice, proj2)} \text{resolvent: } works(g(proj2, alice), proj2) \\
 \frac{\neg supervises(alice, proj2) \ C_7}{\square} \text{resolvent: } supervises(alice, proj2)
 \end{array}$$

Since the empty clause $\square$ was derived, then the negated goal C_8 is refuted; so $\exists S. studies(S, ai)$ holds.